

segment display driver. It can drive up to eight segments. The unique feature of this IC is that it sinks constant current and is not dependent on the supply voltage. So in traffic signals, where the battery voltage varies, the intensity of the display still remains the same.

Software program

The program is written in 'C' language and compiled using Open Source ARDUINO 0022 IDE. It runs smoothly on Linux (Ubuntu), Windows XP as well as MAC OSX, but you may face problems on Fedora and Windows 7.

The source code can be divided into the following parts:

1. Hardware definitions. To make the programming easy, we have defined each pin by its actual function in the circuit along with arbitrary number. For example, the green light signal received at pin 6 of the MCU is named as GREEN_SENSE as follows:

```
#define GREEN_SENSE 4
```

2. Subroutines for display driver TLC5916. The segment data is transferred to TLC5916 between subroutines like Active_output_5916, Inactive_output_5916, Send_byte_5916, Switch2special_5916, Pulse_le, Pulse_clock and Delay_small. Write these subroutines after going through the 5916 datasheet.

3. Timer. We have used timer_ms2 library from Arduino website. It can be set to call a subroutine after a preset time. Here this library calls Set_time every 5 ms.

The following commands are used to implement this function:

```
MsTimer2::set(5, Set_time);
MsTimer2::start();
```

Of these two commands, the first sets the time period to 5 ms and declares the subroutine name as 'Set_time'. The second command starts the timer function.

4. Set_time. Within the Set_time that repeats every 5 ms, the display is refreshed automatically to show the next digit. Set the Half_sec_over

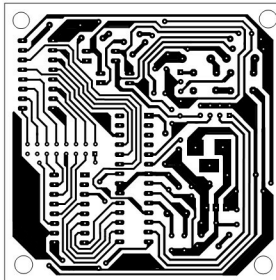


Fig. 3: An actual-size, single-side PCB for the AVR-based traffic light count-down timer

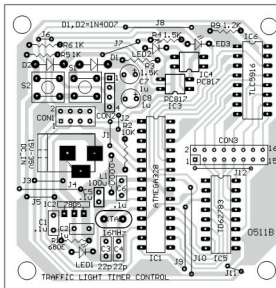


Fig. 4: Component layout for the PCB in Fig. 3

flag when half second is over, i.e., when Set_time is called 100th time. Also, set the One_sec_over flag when one second is over, i.e., when Set_time is called 200th time. At this time, reset the count to zero so as to set Half_sec_over and One_sec_over flags again.

5. Refresh_grid. This function refreshes the display automatically to show only one digit at a time.

6. Setup. This function is called only once at power-on. It sets all the port pins as I/O pins, initialises the

timer and sets the mode as read/scan.

Compile the program

Download Arduino 0022 or the latest version from <http://arduino.cc/> and install like any other program. Now download Mstimer2.zip library file from <http://arduino.cc/en/Reference/Libraries>, unzip and save it in sketchbook/libraries folder.

Start Arduino 0022, load the source file and click Sketch→Verify/Compile or press CNTR-R to compile the program. There should be no error in the program. However, a wrongly placed Mstimer2 library file may cause error.

Burn the program

You can burn the hex code generated by Arduino software in the microcontroller by:

1. Using a third-party universal programmer, or
2. Using an Arduino hardware with bootloader program and burning using Arduino IDE itself. At EFY lab, we used this option to burn our program into the MCU.

This month's EFY-CD contains the hex code file for the traffic timer that will work in ATmega8, ATmega168 and ATmega328. This hex file, along with a standard universal programmer, can be used to program the MCU.

Construction and testing

An actual-size, single-side PCB for the AVR-based traffic light count-down timer is shown in Fig. 3 and its component layout in Fig. 4. An actual-size, single-side PCB layout for